

## UNIT-I

### Definitions

Information Technology: Information technology (or I.T.) is simply defined as the use of computers and software to manage information.

Information Communication Technology (ICT): The term "Information and communication technologies "(ICTs) is defined as a d

Information technology infrastructure: Defined as the shared technology resources that provide the platform for the firms/organiza

The Need for an IT Infrastructure Businesses rely on IS infrastructure to support business processes, decision making and com

Evolution of IT Infrastructure: 1950-2005:

- a) Electronic accounting machine (1930-1950): dominated by machines that began to replace humans in accounting.
- b) General purpose Mainframe and Mini computer Era (1959 – Present): IBM mainframe computing.
- c) Personal computer Era (1981 – Present): Wintel standard and open-source software.
- d) Client / Server Era (1983 – present): Multi-tiered client/server architecture.
- e) Enterprise Computing Era (1992 to present): Internet enterprise-wide networks.
- f) Cloud computing Era (2000 to present): SaaS, IaaS, PaaS utility model.

#### Technological Drivers of Infrastructure Evolution:

- a) Moore's Law and Micro-processing power: Power of microprocessors doubles every 18 months, price falls by half.
- b) Law of Mass Digital Storage: Amount of digital information doubles every year, cost declines 100% per year.
- c) Metcalfe's Law and Network Economies: Value of network grows exponentially as a function of network members.
- d) Declining Communication Costs and the Internet: Rapid decline in internet connection costs.

#### IT/ICT Infrastructure Components:

- Computer hardware platforms: Desktop PCs, laptops, mobile devices, servers.
- Operating System Platforms: Windows, Linux, Unix, macOS.
- Enterprise Software Applications: ERP, CRM, SCM, middleware software.
- Data Management and Storage: SANs, data centers.
- Networking / Telecommunication Platforms.
- Internet platform: Intranets, extranets, Web standards.
- Consulting and System Integration Services: Legacy system integration.

#### Emerging Hardware Platforms and Software Trends:

- Emerging Mobile Digital platform: Smart phones, mobile tablets.
- Grid Computing: Geographically remote virtual supercomputers.
- Cloud Computing: SaaS, IaaS, PaaS model.
- Autonomic Computing: Self-configuring, self-healing systems.
- Edge Computing: Distributed load-balancing near user endpoints.
- Virtualization and Multi-core Processors: VMware server virtualization.

#### Management Challenges:

- Wise infrastructure investments.
- Coordinating hardware, software, CMDB components.
- Building responsive IT infrastructure.
- Scalability and technological change.
- Management and governance.

Role of IT in Business Transformation:

- E-commerce platforms (Amazon elastic cloud infrastructure).
- Real-time mobile banking and fraud detection.
- Electronic Health Record (EHR) systems and telemedicine.
- IoT-enabled predictive maintenance for manufacturing.